

Education

Stony Brook University

Master of Science in Data Science

Incoming Fall 2026

Stony Brook, New York

Experience

Plutous Research

Jan 2026 – Present

Software Engineer

Remote

- Architected **FastAPI** backtesting service on **AWS ECS**; async REST endpoints return job IDs for polling.
- Built **Celery + Redis** task queue for simulation jobs; concurrent execution across **100+** scenarios, **4,000+** steps.
- Built simulation frameworks modeling bid-ask dynamics, inventory risk, and order placement strategies.
- Engineered **PostgreSQL** pipelines to store and evaluate strategies; achieved **~4%** avg ROI, Sharpe **1.8**, Sortino **2.2**.
- Containerized stack with **Docker** on AWS ECS; CloudWatch alerting on failure rates for production observability.

Renaissance School of Medicine, Stony Brook University

Aug 2023 – Dec 2025

Researcher

Stony Brook, New York

- Engineered R pipelines (DESeq2, GSEA) analyzing **20,000+** gene features to map disease progression signatures.
- Built Python analytics pipelines over genomic and clinical data; automated feature extraction and reporting workflows.
- Integrated transcriptomic, histopathology, and spatial imaging into a multimodal feature engineering pipeline.
- Awarded **Snyder Scholar Fellowship** – one of two recipients – securing **\$20,000** in competitive research funding.

Harvard Medical School

Aug 2022 – Dec 2022

Research Assistant

Boston, Massachusetts

- Owned end-to-end build of an AlphaFold/PyMOL pipeline evaluating **100+** gene variants for AAV gene therapy.
- Automated in-silico mutagenesis and stability scoring workflows, reducing manual screening effort by **25%**.

Defence Research and Development Organisation

May 2019 – Aug 2019

Undergraduate Project Intern

Hyderabad, India

- Built a GMDH neural network in Python for antenna direction-finding; reduced RMS error by **67%** (3° to 1°).
- Engineered signal preprocessing pipeline in Python; parsed and normalized raw antenna data for training.

Projects

Protein Intelligence Platform | Python, FastAPI, React, TypeScript, PostgreSQL, Redis, FAISS [🔗](#)

- Built **FAISS** similarity search over high-dimensional protein embeddings; **sub-100ms** latency under load.
- Built a FastAPI-based integration platform aggregating **4** biomedical databases into a unified query interface.
- TypeScript + React frontend with interactive 3D protein visualization; non-technical users explore mutation sites.
- Rate limiter via sorted sets and atomic Lua script; degrades gracefully to passthrough on backend failure.

MemoryOS: Three-Tier Memory Backend for LLM Agents | Python, FastAPI, PostgreSQL, Redis, FAISS, MCP [🔗](#)

- Architected **3-tier** agent memory (Redis, PostgreSQL, FAISS) with importance-scored promotion; **8** REST endpoints.
- Built **384-dim** FAISS retrieval; nightly Celery pipeline clusters and LLM-summarizes episodic memories per agent.
- Exposed **3** MCP tools for native Claude integration; agents autonomously persist and recall context across sessions.
- Containerized **5-service** Docker stack with Alembic migrations and CI; **sub-100ms** end-to-end semantic query latency.

DynaTree: LLM Inference Acceleration via Adaptive Tree Speculative Decoding | Python, PyTorch

- **Training-free** speculative decoding with confidence-aware dynamic tree structure; lossless via single-pass tree attention.
- Outperformed EAGLE-2 by **7.4%** on WikiText-2; achieved **9.4×** speedup over autoregressive decoding.
- Dynamically adjusts tree breadth and depth per token; no draft model training or architecture changes required.

Skills

Languages: Python, C++, TypeScript/JavaScript, SQL, R, MATLAB

Full-Stack & Infra: React, FastAPI, PostgreSQL, Redis, Docker, AWS, GCP, Git, Unix/Linux

Software Engineering: OOP, REST APIs, Distributed Systems, Async Processing, Microservices

Data & ML: Pandas, NumPy, PySpark, PyTorch, Scikit-learn, XGBoost, LangChain, HuggingFace, FAISS

Publications

- Differentiable linear dynamical layers for structured deep learning in sequential modeling, 2026 [🔗](#)
- Advancing exchange liquidity: Unveiling the evolution of market makers, 2026 [🔗](#)
- Predicting stock price movements in high-frequency trading, 2026 [🔗](#)
- DNA-PKcs/JNK/p53 pathway underlies cell fate decisions during DNA replication catastrophe, 2025 [🔗](#)